(a) Spatial Correlation
Ensemble-mean composites vs EOF modes

	EOF 1	EOF 2	EOF 3
Cluster 1 (Low-amplitude SAM−)	**-0.95** ±0.57	**-0.01** ±0.18	**-0.00** ±0.14
Cluster 2 (SAM+)	**+0.98** ±0.01	**+0.14** ±0.09	**-0.04** ±0.05
Cluster 3 (SAM−)	**-0.98** ±0.03	**-0.16** ±0.09	**+0.04** ±0.08

(b) Temporal Correlation
Direct cluster projections vs PCs

	PC 1	PC 2	PC 3
Cluster 1 (Low-amplitude SAM−)	**-0.99** ±0.59	**-0.01** ±0.13	**+0.02** ±0.10
Cluster 2 (SAM+)	**+0.99** ±0.00	**+0.09** ±0.06	**-0.04** ±0.03
Cluster 3 (SAM−)	**-0.99** ±0.01	**-0.11** ±0.06	**+0.04** ±0.05

−1.0 −0.5 0.0 0.5 1.0

Pearson r

Large value: correlation of the 10-member ensemble-mean composite. Small value: standard deviation across the 10 members computed individually.